BUDESONIDE

Nebulised in croup

High first-pass metabolism

CHEMICAL NAME (RS)-11β,21-dihydroxy-16α,17α-(butylidenebis(oxy))pregna-1,4-diene-3,20-dione

CLASS Corticosteroid (inhaled, nebulised, nasal, oral gut-targeted)

TRADE NAMES Pulmicort · Rhinocort · Entocort / Budenofalk (oral) · Symbicort (with formoterol)

THERAPEUTIC USES / INDICATIONS

- Asthma maintenance (inhaled/nebulised, incl. young children).
- Croup (nebulised, alternative to oral dexamethasone).
- Allergic rhinitis, nasal polyps (nasal).
- Crohn's disease (ileocaecal), microscopic colitis (oral MR); ulcerative colitis (rectal foam/enema).

ADMINISTRATION

- **Inhaled** (Turbohaler) twice daily; rinse mouth after.
- **Nebulised** suspension via jet nebuliser (not ultrasonic) with mask/mouthpiece; wash face after mask.
- **Oral MR capsules** swallowed whole in the morning.

DOSAGES

- **Asthma (inhaled):** 200–400 mcg twice daily (max 1,600 mcg/day).
- **Nebulised asthma:** 0.5–1 mg twice daily (child 0.25–0.5 mg).
- **Croup:** 2 mg nebulised single dose.
- **Crohn's:** 9 mg once daily for up to 8 weeks, then taper.
- **Nasal (OTC spray):** adults and ≥12 yr 2 sprays each nostril once daily, reduce to 1 spray when controlled.

PHARMACOKINETICS

- **Onset:** inhaled days–weeks; croup 30 min–2 h.
- ~90% first-pass hepatic metabolism (CYP3A4) → low systemic effects (basis for oral gut use).
- Excreted in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Potent **glucocorticoid** → local anti-inflammatory effect in airways or gut mucosa.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Oral thrush, hoarseness (inhaled).
- Nasal irritation, epistaxis.
- Oral: dyspepsia, headache.

ADVERSE EFFECTS (SERIOUS)

- Systemic steroid effects at high doses (adrenal suppression, growth, bone).
- Facial skin irritation from nebuliser mask.
- Oral: Cushingoid features (less than prednisolone).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** TB, infections, liver cirrhosis (oral; ↑ systemic exposure).

KEY INTERACTIONS

- **Ketoconazole, itraconazole, ritonavir, grapefruit juice:** ↑ systemic levels.

NURSING MANAGEMENT

- Check inhaler/nebuliser technique; wash face after mask.
- Croup: monitor stridor, recession, SpO_2, agitation.
- Monitor for thrush.
- Crohn's: assess stool frequency, pain.

PATIENT EDUCATION

- Rinse mouth after inhaling.
- Use daily for prevention.
- Oral capsules: take in the morning, swallow whole.
- Avoid grapefruit juice.

KEY POINTS

- Versatile steroid: inhaled, nebulised, nasal, gut.
- High first-pass → fewer systemic effects.
- Croup 2 mg nebulised.
- Rinse mouth.

SOURCES CHECKED **[466]** Budesonide 0.5mg nebuliser suspension (UK SmPC) **[467]** Pulmicort Turbohaler 200 (UK SmPC) **[468]** budesonide, oral (US label) **[469]** budesonide, oral (US label, Major Pharmaceuticals)

Full references p. 432–457